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Rotor Dynamics Tools Analysis
Computer Aided Engineering - Project 1

17.10.2025

1 Introduction

The use of Computer-Aided-Engineering (CAE) has become essential in modern structural analysis. It allows engineers to model and evaluate complex dynamic responses. Computer-Aided-Softwares like SolidWorks, Ansys and Matlab, provide the real-world conditions that can be accurately simulated using real-life conditions and boundaries. In this study: Project Part 1, the contact nonlinearity between two beams made of two different materials: steel and aluminum is studied. The parts are modeled in SolidWorks with a 90 mm overlap. The steady-stated thermal study is conducted with refined meshing and boundary conditions in Ansys. The resulting model was then examined in Matlab using eigenvalues and frequency analysis to depict natural frequencies and mode shapes which are important for predicting resonance and deformation patterns.
This project emphasizes the importance of accurate modeling, meshing and boundary condition definitions as well as in obtaining, interpreting and selecting the reliable results. This study aim to enhance the understanding of nonlinear behavior in beam structure.

2 Methodology

2.1 Softwares

The project was analyzed using three different computer aided software: SolidWorks, Ansys, and Matlab.
SolidWorks: This computer aided design was used to model the beams as individual parts. Then the designed beams are assembled in the Assembly feature by applying the mate function to position them as actual design using the respective geometries constraints and dimensions.
Ansys: With the help of Mechanical feature of this workbench, a nonlinear analysis was conducted by using a steady-state thermal study. It was performed by defining the mesh, specifying the contact points and nodes, and applying fixed constraints and required temperatures.
Matlab: Numerical analysis was conducted in Matlab by defining the properties of the material and calculating various graphical representations and eigenmodes by evaluating the eigenvalues.

2.2 SolidWorks - Modeling & Assembling

Firstly, the bodies that are required to analyze were fabricated in the computer aided design (CAD) in Solidworks's part feature. The dimensions of the parts are shown below in Figure 1. The defined square beam is extruded by 600 mm.
The model parts are imported into SolidWorks Assembly. The beams are mated using the respective geometries features to place in the desired position. The two beams are stacked onto each other and the offset distance between the two faces is 90 mm as shown in Figure 4

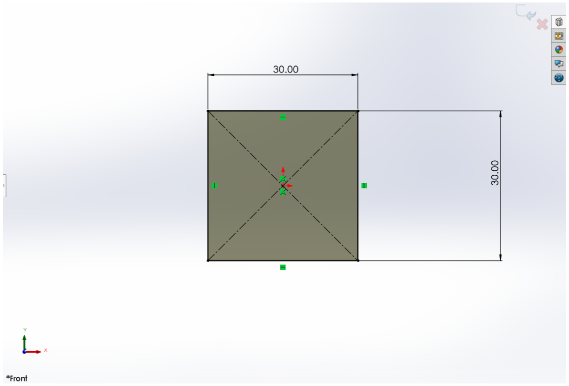

Figure 1: Dimension of the beam

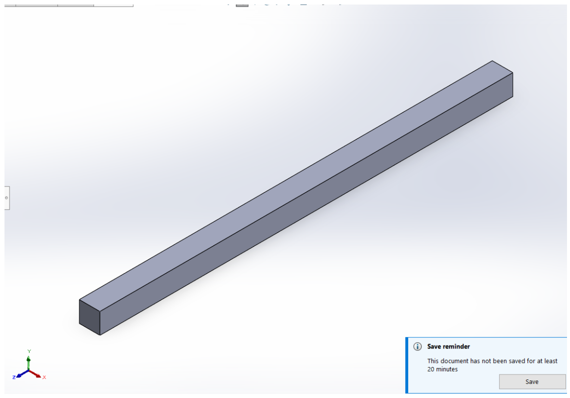

Figure 2: Extruded Beam

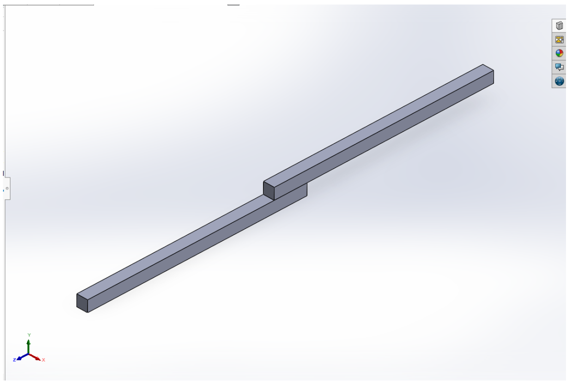

Figure 3: Isometric view of the Assembly

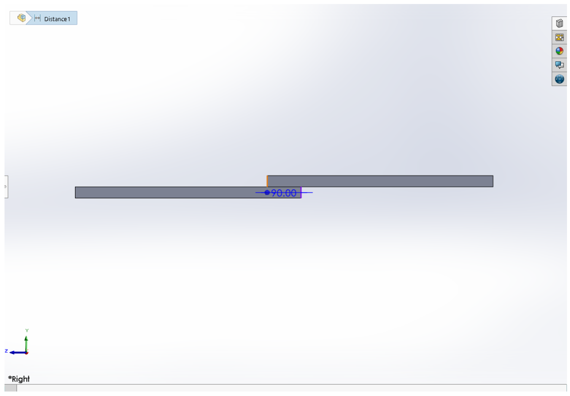

Figure 4: Right view of the Assembly

The desired coordinate system is created by placing at the corner of the lower beam. The purpose of establishing the new coordinate is to ensure that subsequent analysis is oriented consistently. The desired coordinate is Y axis points vertically, X axis points horizontally and Z axis is directed outward from the screen. Then, the assembly is exported as STP file with the new coordinate system in the output coordinate so it can be analyzed in Ansys software. The parts were also made sure it was in the correct dimensions: MKS (millimeter, kilogram, second).

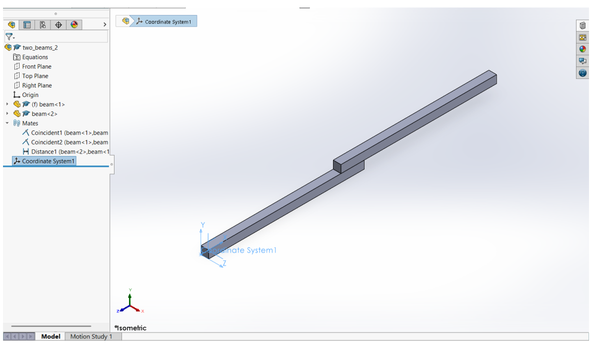

Figure 5: New desired coordinate system establishment

2.3 Ansys Workbench & Mechanical

The modeled the stp file is imported into Workbench of Ansys. The steady-state thermal analysis was inserted into the model.

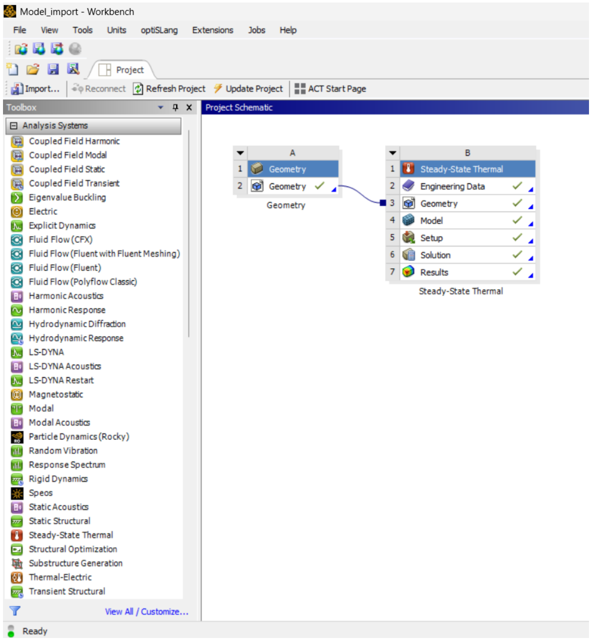

Figure 6: Workbench Interface: Steady-state Thermal

In the Ansys mechanical program, the mesh of the beams are defined using the tetrahedrons. The mesh are refined repeatedly until the the desired outcome has been meshed to the point that there are three nodes at each edge of the beam. The desired mesh was set at $2.75e^{-002}$. The edges of the beams are then grouped into a selection group and, it was filtered into nodes only. The selection group is named as ALL_SURFACES so it could match with the function name in Matlab.

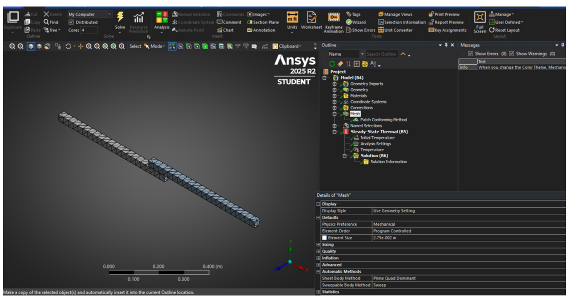

Figure 7: Mechanical Interface: Meshing was updated

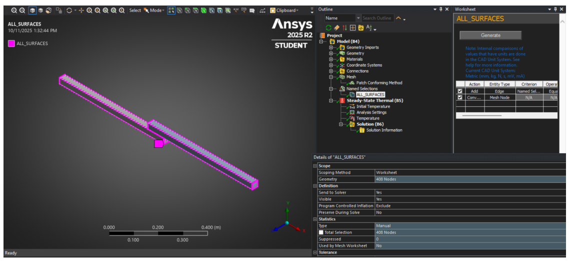

Figure 8: Named Selection Group: ALL_SURFACES

The temperature was inserted on the right face of the top beam with 22 °C as shown in the Figure 9. After everything was set up, the simulation was solved and saved a project.

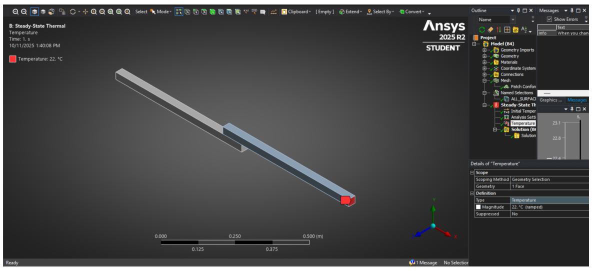

Figure 9: Setting the temperature on the surface of a beam

2.4 Matlab

2.4.1 File Preparation

The ready made Matlab zip file was downloaded from the assignment and unpacked. First the file preparation for Matlab has to be arranged by extracting "ds.dat" file from Ansys Project File into Matlab directory folder where main_two_beams.m file is located. As there was an original "ds.dat" initially there, the initial file was renamed to "ds_old.dat". The main_two_beams.m file was initiated by using the initializePaths.m so that it can run on the Matlab.

2.4.2 Materials Properties Setting

Two materials are assigned at each beam. The materials are Steel & Aluminum.
Steel:

- Young Modulus (E) = $200e^9$
- Poisson Ratio (v) = 0.3
- Density (ρ) = 7850

Aluminum:

- Young Modulus (E) = $68e^9$
- Poisson Ratio (v) = 0.33
- Density (ρ) = 2770

2.4.3 Computing

The Rotor Dynamic 3D package was analyzed using the material properties to obtain eigenvalue parameters such as mass matrix ($\mathbf{M}$), stiffness matrix ($\mathbf{K}$) and gyroscopic matrix ($\mathbf{G}$) of each beam. The eigenmodes of the beams are visualized through graphical analysis. The results were then analyzed and examined to evaluate for its vibrational modes with their corresponding frequencies. The eigenmode with zero vibrations are discarded.

3 Results

Matlab utilizes the information from the model with constraints information provided from Ansys file (ds.dat) from the project folder and it provides the depiction of the models and relative deformation as eigenmodes, varying frequencies in Hertz (**Hz**). As shown in Figure 10, the first two pictures are the illustration of the beam bodies. Body 1 and 2 are assigned to Steel & Aluminum and their property respectively.

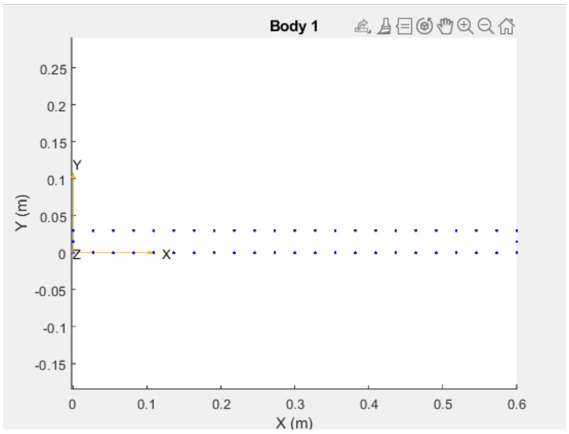

Figure 10: BODY 1

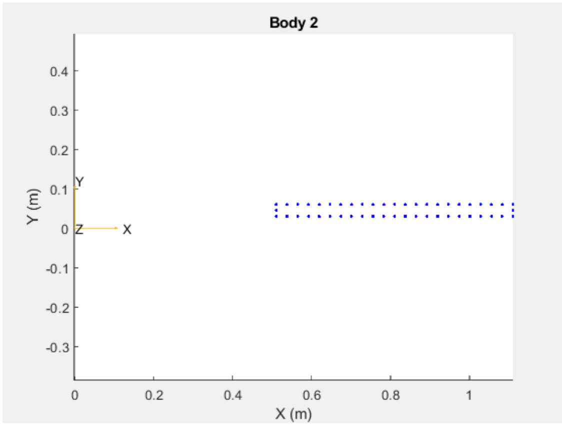

Figure 11: BODY 2

Initially, the code was instructed with eigenmodes (n) from 1 to 10. The eigenmodes with zero frequencies are discarded as they do not contribute to the dynamic behavior. The first non-zero result appeared at the sixth eigenmode with $2.8342e^{-4}$. The analysis was repeated with eigenmode n = 6 to 20. The results are correctly oriented, confirming that the applied load acted along the negative Y-axis direction. The analysis that are relevant to the accuracy of the dynamic behavior were selected to analyze furthermore.

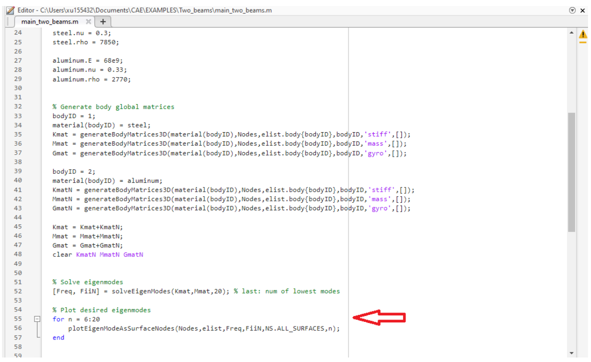

Figure 12: Matlab code iteration

Eigenvalues represent the critical vibration speed of the structure, indicating the specific frequencies at which the structure tends to vibrate. The corresponding eigenvectors illustrate how the structure deforms at those frequencies. Generally, higher the eigenmodes correspond to higher frequency value and more localized deformation.

A few representation of the results are provided below, while a detailed analysis is presented in the following section.

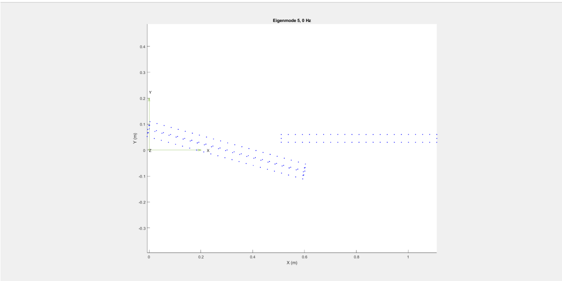

Figure 13: Eigenmode 5

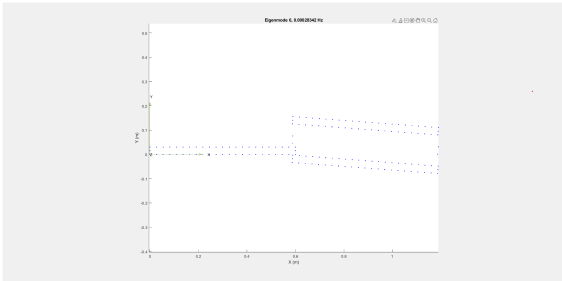

Figure 14: Eigenmode 6

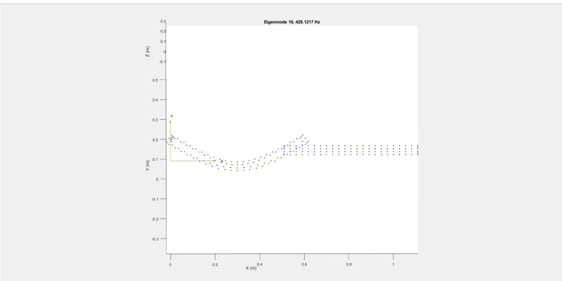

Figure 15: Eigenmode 16

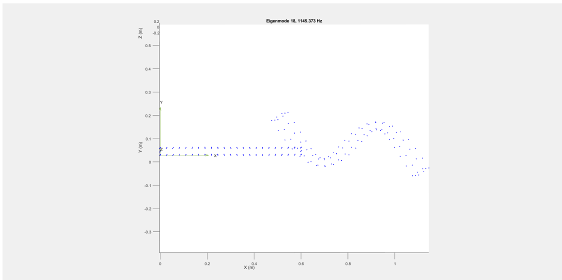

Figure 16: Eigenmode 18

4 Analysis

4.1 Criteria for Reliable Results

The analysis was conducted with eliminating eigenmodes that are not useful for the accuracy of the dynamic analysis. To achieve an accurate and meaningful result:

- The displacement along negative Y-axis were considered in alignment with expected loading condition.
- The representation of the beam's geometry and dimension but in the higher eigenmodes, any displacement along Y-axis were accounted for, as they may be relevant for the future analysis.
- Additionally, the realism of the computed frequencies was taken into account as well.

4.2 Eigenmode Selection

The mode shapes that are removed that were not met above criteria are shown below. The first figure shows that the body of the beam does not represent the physical body of the beam. the second figure depicts the movement on X-Z plane and it may not be very useful for the accuracy Figure 17.

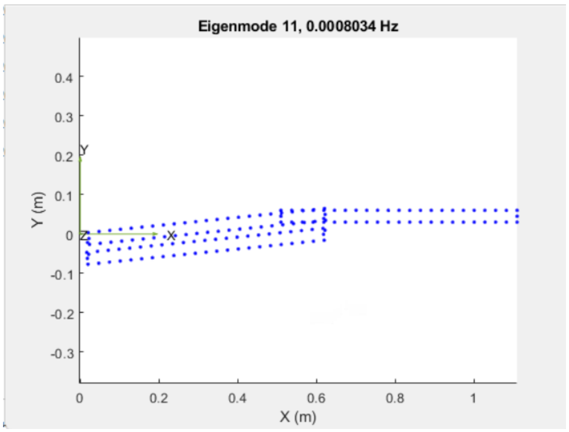

Figure 17: Eigenmode 11

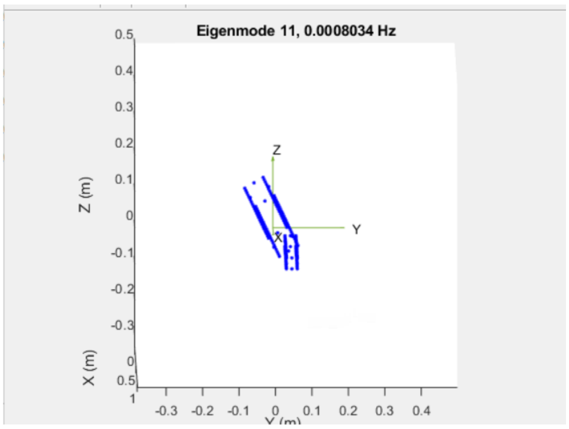

Figure 18: Eigenmode 11: Displacement on X-Z Plane

The good example of the following criteria is in Figure 19. As the contact point was under vertical load, the relative deformation was also in Y-axis with rotational effect on beam 2.

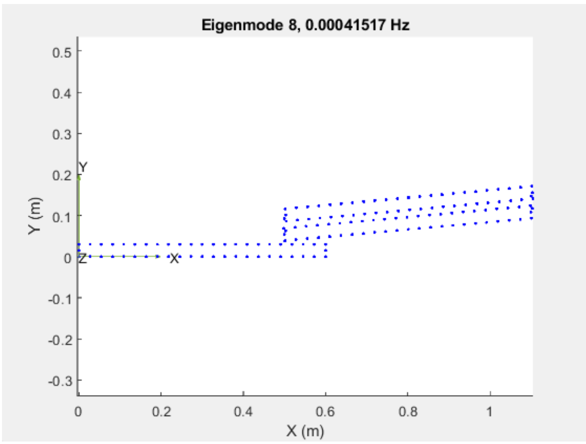

Figure 19: Eigenmode 8

The reliability of the results depend on the selecting of the figures that are relevant to the results. Figure 20 show that in Eigenmode 13 and 14, Eigenmode 14 is more reliable due to its deformation in Y-axis.

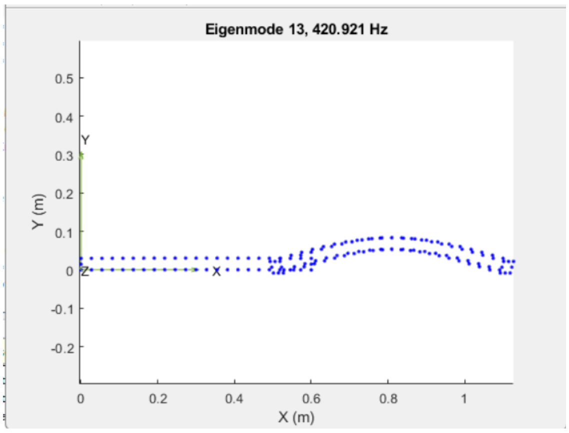

Figure 20: Eigenmode 13

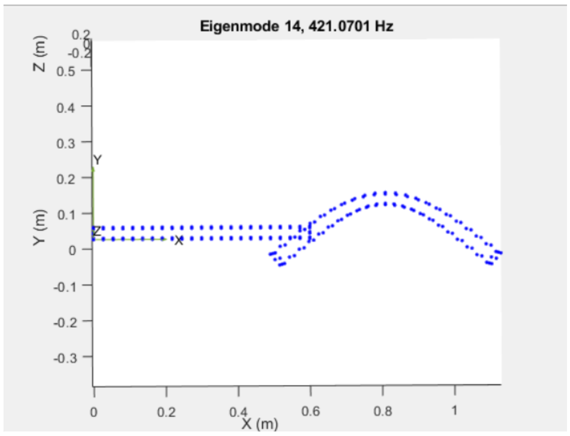

Figure 21: Eigenmode 14

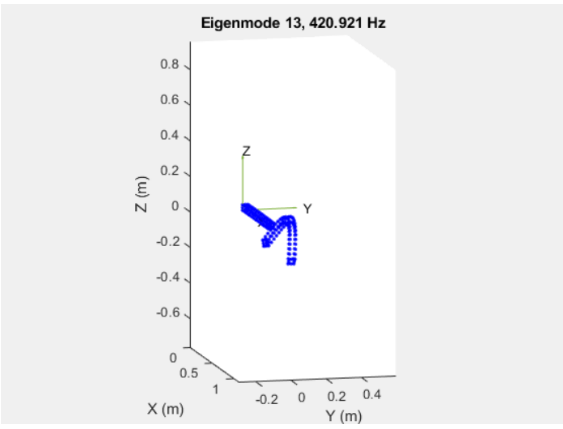

Figure 22: Eigenmode 13 Rotated view

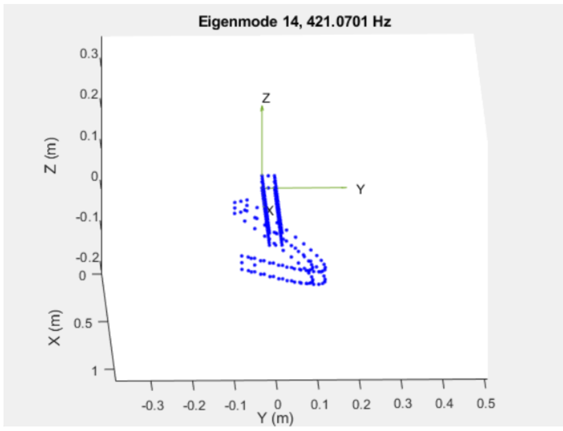

Figure 23: Eigenmode 14 Rotated view

It is also observed that material of the beams are crucial in the study. The aluminum beam deforms at lower frequencies at 1144.0412 **Hz**, whereas the steel beam deforms at higher frequencies at **Hz**. This difference is primarily due to material properties: aluminum having lesser density and lower stiffness compared to steel.

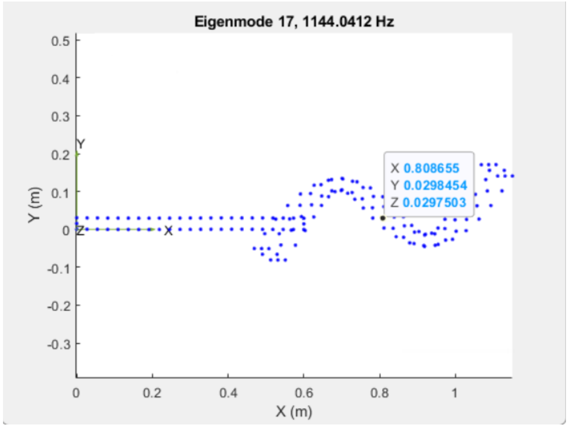

Figure 24: Eigenmode 17: Aluminum

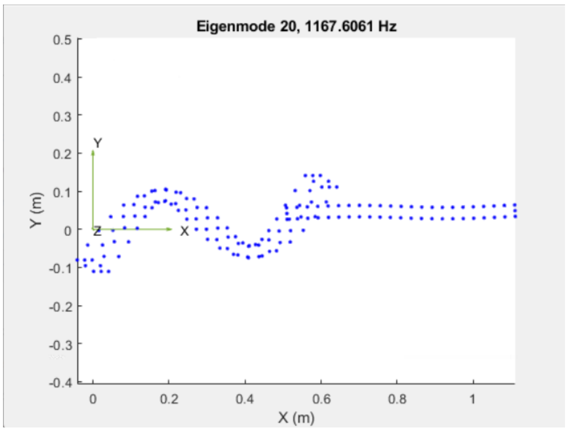

Figure 25: Eigenmode 20: Steel

4.3 Summary of Results

The eigenmode analysis provided valuable insight into the dynamic characteristics of the beam structure. The observed deformation patterns confirmed that the structure initially behaves as a system under vertical loading.

- The elimination of irrelevant eigenmodes improve the accuracy of the dynamic analysis.
- Aluminum having lower density compared to steel, resulted in higher natural frequencies due to its lower stiffness-to-weight ratio. The impact of material can have significant impact on the results.
- The nodal results provided the relative displacement of each point in the structure, allowing the prediction of the physical deformation pattern.
- The eigenmode analysis identifies critical speeds (vibrational speed) at which the beam could experience significant deflection or resonance.
- Patterns of deformation can be estimated based on material properties and boundary conditions such as applied forces and environmental variables like temperature.

5 Discussion

5.1 Challenges & Key Takeaways

The study provides me valuable insights into the process of conducting a structural analysis from designing a model to confining the constraints to numerical analysis through various softwares: SolidWorks, Ansys and Matlab. Although I was already familiar with all of the softwares tools. I spent significant amount of time trying to get the same results because of the different versions of the softwares which leads to

technical problems. The same package file with different version of Ansys and Matlab, can provide different results.
Despite the technical issues, the task offered contains insightful lessons and knowledge on how a non-linear study can be conducted. It highlights the importance of selection of proper eigenmode for the accurate results, meshing strategy, the boundary conditions in ensuring the reliability of the study.

5.2 Limitation & Recommendations

As the instructor stated in the video, there are many factors that can limit computing the numerical simulation to mimic the real-world problem. The manufacturing defects, geometric imperfections and the material variation can limit the characteristics of the model. The current analysis did not account for non-linearity effects which may lead to nonlinear responses in practical conditions.
For future studies, it is recommended to incorporate transient temperature and load over time to simulate the non-linearity effect. The future study on the effect of temperature on deformation and stress distribution would provide a more realistic representation of operational environment. The boundaries conditions such as friction and damping effect in model can provide more reliable and accurate results.